



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	THERMODYNAMICS				
Type of Course	Core in Major				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Proficiency in calculus, basic physics principles including mechanics and heat transfer, and a foundational understanding of chemistry are typically prerequisites for an undergraduate thermodynamics course.				
Course Summary	Thermodynamics course covers fundamental principles such as the conservation of energy, entropy, and thermodynamic properties of substances, providing students with the knowledge to analyse and predict the behaviour of systems in various contexts, from power generation to environmental processes				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate a solid understanding of the fundamental principles of	Understanding	Fundamental Principles	Conceptual quizzes,

	thermodynamics, including the laws of thermodynamics and their mathematical representations.			written examinations
CO2	Apply thermodynamic concepts to analyze and solve problems in classical physics.	Application	Application in Classical Physics	Problem-solving exercises, laboratory experiments
CO3	Utilize mathematical tools, including calculus and differential equations, to model thermodynamic systems and predict their behavior.	Application	Mathematical Modelling; Evaluation Tools	Mathematical problem sets, computational assignments
CO4	Interpret thermodynamic properties of materials and their phase transitions, connecting theoretical concepts with experimental observations.	Analysis	Knowledge Category: Properties of Matter	Data analysis projects, laboratory reports
CO5	Evaluate and compare the efficiency and performance of thermodynamic processes and cycles, including practical applications such as heat engines and refrigeration systems.	Evaluation	Efficiency and Performance	Performance assessments, design projects
CO6	Apply thermodynamics principles to interdisciplinary areas such as materials science, environmental science, and astrophysics, demonstrating the relevance and versatility of thermodynamic concepts.	Application	Interdisciplinary Applications	Research projects, case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	ZEROth LAW AND FIRST LAW OF THERMODYNAMICS		10	16
	1	Thermodynamic Limit, Extensive and Intensive Variables, The Ideal Gas	1	
	2	A Definition of Heat, Heat Capacity, Specific Heat Capacity, Molar Heat Capacity, C_p and C_v	1	
	3	Thermal Equilibrium and Zeroth Law of Thermodynamics	1	

	4	A System in Thermal Equilibrium, Functions of State	1	
	5	The First Law of Thermodynamics	1	
	6	Heat Capacity	2	
	7	Reversibility	1	
	8	Isothermal and Adiabatic Expansion of Ideal Gas	2	
	Sections 1.2 – 1.3 of chapter 1, 2.1 – 2.2 of chapter 2, 4.1 of chapter 4, 11.1 – 11.3 of chapter 11, 12.1 – 12.3 of chapter 12 of Book 1			
II	HEAT ENGINES AND THE SECOND LAW OF THERMODYNAMICS		9	16
	9	The Second Law of Thermodynamics	1	
	10	The Carnot Engine	2	
	11	Carnot's Theorem	1	
	12	Equivalence of Clausius and Kelvin Statements	1	
	13	Examples of Heat Engines	1	
	14	Heat Engines Running Backwards: Refrigerator and Heat Pump	1	
	15	Clausius' Theorem	2	
	Sections 13.1 – 13.7 of chapter 13 of Book 1			
III	ENTROPY		10	16
	16	Definition of Entropy	1	
	13	Irreversible Change	2	
	14	The First Law Revisited	1	
	15	The Joule Expansion of Ideal Gas	1	
	16	The Statistical Basis for Entropy	1	
	17	Entropy of Mixing	1	
	18	Maxwell's Demon	1	
	19	Entropy and Probability	2	

	Sections 14.1 – 14.8 of chapter 14 of Book 1			
IV	THERMODYNAMIC POTENTIALS, THIRD LAW OF THERMODYNAMICS AND PHASE TRANSITIONS		16	22
	20	Thermodynamic potentials – Internal Energy (U), Enthalpy (H), Helmholtz Function (F) and Gibbs Function (G)	3	
	21	Availability and Constraints	1	
	22	Maxwell's Relations	2	
	23	Different Statements of the Third law and the Consequences of the Third Law	2	
	24	Latent Heat, Clausius – Clapeyron Equation and Phase Diagrams	3	
	25	Stability and Metastability	2	
	26	Gibbs Phase Rule	1	
	27	Classification of Phase Transitions	2	
	Sections 16.1 – 16.6 of chapter 16, 18.1 – 18.2 of chapter 18, 27.1 – 27.3 of chapter 27, 28.1 – 28.5, 28.7 of chapter 28 of Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Verification of Boyle's law and Charle's law - Boyle's law ($PV = \text{a constant}$) states that at a constant temperature, volume of a gas is inversely proportional to pressure. - Determine the volume - pressure relation at constant temperature using the water column. - Plot the pressure versus volume graph and verify Boyle's law. - Verify the law at minimum two different temperatures. - Charle's law ($V/T = \text{a constant}$) states that at constant pressure, volume is directly proportional to temperature. - In this experiment determine the temperature - volume relation at constant pressure using the water column.		

		- Plot the temperature versus volume graph and verify the Charle's law. - Verify the law at minimum two different pressures.		
	2	Verification of Gay-Lussac's law - Gay-Lussac's law ($P/T = \text{a constant}$) states that at constant volume, pressure is directly proportional to temperature. - In this experiment determine the temperature - pressure relation at constant pressure using metallic bulb and water column or pressure gauge or using Jolly's bulb apparatus. - Plot the temperature versus volume graph and verify the Charle's law.		
	3	Specific heat of metal - Specific heat of an object is the amount of heat required to change the temperature by unit degree Celsius per unit mass. - The amount of heat transferred and the change in temperature can be obtained using suitable metal object and water bath. - The metal block of suitable mass is kept in constant temperature water bath at a higher temperature/boiling water, until thermal equilibrium is attained. Then immerse the metal block in a beaker filled with water at room/lower temperature until thermal equilibrium is attained. - By equating heat gain (of water and beaker) to heat loss (of metal rod) and the temperature change of metal block, specific heat of metal can be estimated. - Determine the specific heat of at least two different metals.		
	4	Latent heat of fusion of ice - Latent heat of ice is the heat energy absorbed to change its phase from solid to liquid without changing its temperature. - To measure the heat transferred, cubes of ice are mixed in water taken in a beaker. - By equating heat gain by (ice to melt and melt ice to rise its temperature up) to heat loss (by water and beaker), the latent heat of fusion of ice can be determined. - Experiment should be performed in thermally insulated / thermocol box.		
	5	Thermal conductivity by Searle's method		

		Determine the thermal conductivity of copper or any other metal using Searle's method / apparatus.		
	6	Thermal conductivity by Forbes method Determine the thermal conductivity of steel or copper or any other metal using Forbes method.		
	7	Temperature coefficient of resistance of a metal - Resistance of metals increases with increase in temperature. - Measure the resistance of the metal coil, using Carey Foster's bridge or Potentiometer or any other suitable method, as a function of temperature from 100 degree Celsius to room temperature. - Plot graph and find the temperature coefficient of resistance.		
	8	Characteristics of NTC thermistor - Resistance of Negative Temperature Coefficient (NTC) thermistors decreases with increase in temperature. - Measure the resistance of the thermistor, using Carey Foster's bridge or Potentiometer or ExpEYES or any other suitable method, as a function of temperature from 100 degree Celsius to room temperature. - Plot the graph and study the characteristics.		
	9	Band gap of a semiconductor - Measure the reverse bias current/resistance of a semiconductor diode as a function of temperature, using Carey Foster's bridge or Potentiometer or ExpEYES or any other suitable method. - Plot the logarithm of resistance/current against the inverse of temperature. - From the slope, the band gap from the semiconductor can be obtained.		
	10	Thermo emf of a Thermocouple Study the variation of thermo emf of a thermocouple as a function of temperature of the hot junction while maintaining the cold junction at 0 degree Celsius.		

11	Newton's law of cooling - According to Newton's law of cooling, the rate of heat loss of a hot body is proportional to the difference in temperature between the body and the surroundings. - The calorimeter is filled with hot water and the variation in temperature is noted as a function of time. - Cooling rate graph is plotted and law is verified. - Emissivity of the surface of the calorimeter can also be determined. - ExpEYES and PT1000 sensor may be used to record the temperature. https://expeyes.in/experiments/thermal/cooling.html		
12	Thermal conductivity of a bad conductor by Lee's Disc method Determine the thermal conductivity of a bad thermal conductor using Lee's disc apparatus.		
13	Determination of coefficient of linear thermal expansion of metal - Linear coefficient of thermal expansion is the change in length of a material per unit change in temperature per unit length. - Measure the length of a long metal rod as function of temperature. - Plot the length / change in length of the rod as function of temperature. - From the slope coefficient of linear thermal expansion of metal can be obtained. - Perform the experiment for minimum two different metals.		
14	Melting point of wax - Fill a test tube with wax until half and use a thermometer inside the wax / test tube to measure wax temperature. Avoid the thermometer touching the test tube. - Immerse the test tube in a water bath with the help of a stand, in such a way that the wax is below the water level. - Use a suitable flame / heating rate and measure the wax temperature as a function of time at a suitable time interval. - Plot temperature versus time graph. ExpEYES and PT1000 sensor may be used to record the temperature. https://expeyes.in/experiments/thermal/cooling.html - The temperature increases initially and remains constant until the wax melts completely. The flat temperature gives		

		the melting point of wax (The melting point depends on the type of wax used)		
	15	Simulate the Thermodynamic process in PV diagram and estimate the work done by numerical integration - Plot isothermal, adiabatic and isobaric process in the PV diagram. - Estimate the work done by numerical integration in each case. - Refer section 4.2 of Book 2		
	16	Simulate the Carnot Cycle in PV diagram and estimate the efficiency by numerical integration - Plot the Carnot cycle in the PV diagram. - Estimate the work done in each process by numerical integration and estimate efficiency. - Compare the estimated efficiency with theoretical efficiency. - Refer section 4.6 of Book 2		

Books and References:

1. Concepts in Thermal Physics by Stephen J Blundell and Katherine M. Blundell (Book 1)
2. Thermal Physics Tutorials with Python Simulations (CRC Press, 2023) by Minjoon Kouh and Taejoon Kouh (Book 2)
3. Thermal Physics by Charles Kittel and Herbert Kroemer
4. An Introduction to Thermal Physics by Daniel V. Schroeder
5. Heat and Thermodynamics by Mark Zemansky, Richard Dittman
6. Thermal Physics by Garg, Bansal, and Ghosh
7. Thermodynamics and Statistical Physics by Satya Prakash
8. Heat Thermodynamics and Statistical Physics by Brij Lal , N Subrahmanyam and PS Hemne
9. NPTEL video lectures: <https://nptel.ac.in/courses/115106090>

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	0	0	0	0	0	1	0	0	1	0	0	0
CO 2	2	0	3	0	2	0	1	0	0	1	0	0	0
CO 3	2	0	3	0	2	0	0	0	0	1	0	0	0

CO 4	2	0	2	0	2	0	0	0	0	1	0	0	0
CO 5	2	0	3	0	2	0	0	0	0	1	0	0	0
CO 6	2	0	3	0	3	0	0	0	0	1	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	